Mini project report on

DECENTRALIZED ESCROW & MULTI-SIGNATURE PAYMENT SETTLEMENT ENGINE

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE24CS351A – DBMS Project

Submitted by:

R Muralidharan	**PES2UG24CS387**
Punith K	**PES2UG24CS383**

under the guidance of

Prof Vinu J

Assistant Professor

PES University

AUG - DEC 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

CERTIFICATE

This is to certify that the mini project entitled

Disasters data Management System

is a bonafide work carried out by

R Muralidharan	**PES2UG24CS387**
Punith K	**PES2UG24CS383**

In partial fulfilment for the completion of fifth semester DBMS Project (UE24CSS351A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Prof Vinu J
Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **DECENTRALIZED ESCROW & MULTI-SIGNATURE PAYMENT SETTLEMENT ENGINE** has been carried out by us under the guidance of **Prof Vinu J, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

R Muralidharan	**PES2UG24CS387**
Punith K	**PES2UG24CS383**

ACKNOWLEDGEMENT

I would like to express my gratitude to Prof Vinu J, Department of Computer Science and Engineering, PES University, for her continuous guidance, assistance, and encouragement throughout the development of this UE24CS351A - DBMS Project.

I take this opportunity to thank Dr. Sandesh B J, C, Professor, ChairPerson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department.

I am deeply grateful to Prof. Jawahar Doreswamy, Chancellor – PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University for providing to me various opportunities and enlightenment every step of the way. Finally, this DBMS Project could not have been completed without the continual support and encouragement I have received from my family and friends.

ABSTRACT

The Decentralized Escrow & Multi-Signature Payment Settlement Engine is a database-driven application designed to provide secure and transparent payment settlement between buyers and sellers. In conventional online transactions, users may face challenges related to payment security, trust, unauthorized fund release and dispute resolution. This project addresses these challenges through an escrow-based payment management system that ensures funds are released only after the required approval conditions are satisfied.

The system uses an escrow mechanism in which the buyer's payment is associated with a transaction and remains under controlled settlement until the predefined conditions are met. A multi-signature approval mechanism is implemented through configurable signing rules, allowing multiple authorized participants to approve a transaction before funds can be released. The application supports different user roles, including Buyer, Seller and Arbitrator.

The relational database is implemented using MySQL 8 and consists of seven primary tables: Users, Multisig Rules, Escrows, Escrow Signers, Approvals, Transactions and Disputes. These tables maintain user information, escrow details, approval requirements, transaction records and dispute information while enforcing primary keys, foreign keys, unique constraints and check constraints.

The system also supports transaction lifecycle management, including escrow creation, approval collection, fund release, dispute handling, refunds and expiry management. Database programming features such as stored procedures, functions and triggers are used to implement business rules and maintain data consistency. MongoDB can additionally be used for maintaining audit events and recording important system activities, where implemented.

A web-based frontend developed using Next.js provides an interface for users to interact with the system. The application integrates with the database to provide transaction visibility, approval tracking and dispute management.

The project demonstrates the practical application of relational database design, normalization, integrity constraints, SQL queries, database programming and frontend-database integration in developing a secure payment settlement management system.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	11
2.	PROBLEM DEFINITION	12
3.	ER MODEL	13
4.	ER TO RELATIONAL MAPPING	14
5.	DDL STATEMENTS	15
6.	DML STATEMENTS	20
7.	QUERIES (SIMPLE QUERY AND UPDATE AND DELETE OPERATION, CORRELATED QUERY AND NESTED QUERY)	23
8.	STORED PROCEDURE, FUNCTIONS AND TRIGGERS	30
9.	FRONT END DEVELOPMENT	40
REFERENCES/BIBLIOGRAPHY		41
APPENDIX A DEFINITIONS, ACRONYMS AND ABBREVIATIONS		42

LIST OF TABLES

Table No.	Title	Page No.
3.1	Description of Entities in the ER Model	XX
3.2	Entity Relationships and Cardinalities	XX
4.1	Relational Schema of the Proposed System	XX
5.1	Users Table Structure	XX
5.2	Multisig Rules Table Structure	XX
5.3	Escrows Table Structure	XX
5.4	Escrow Signers Table Structure	XX
5.5	Approvals Table Structure	XX
5.6	Transactions Table Structure	XX
5.7	Disputes Table Structure	XX
5.8	Database Constraints	XX
6.1	Sample User Records	XX
6.2	Sample Multisig Rules Records	XX
6.3	Sample Escrow Records	XX
6.4	Sample Escrow Signers Records	XX
6.5	Sample Approval Records	XX
6.6	Sample Transaction Records	XX
6.7	Sample Dispute Records	XX
7.1	Aggregate Query Results	XX
7.2	Correlated Query Results	XX
7.3	Nested Query Results	XX

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	Entity Relationship Diagram	XX
4.1	Complete ER to Relational Mapping Diagram	XX
5.1	Database Creation	XX
5.2	Users Table Creation	XX
5.3	Multisig Rules Table Creation	XX
5.4	Escrows Table Creation	XX
5.5	Escrow Signers Table Creation	XX
5.6	Approvals Table Creation	XX
5.7	Transactions Table Creation	XX
5.8	Disputes Table Creation	XX
6.1	Insertion of User Records	XX
6.2	Insertion of Multisig Rules Records	XX
6.3	Insertion of Escrow Records	XX
6.4	Insertion of Escrow Signers Records	XX
6.5	Insertion of Approval Records	XX
6.6	Insertion of Transaction Records	XX
6.7	Insertion of Dispute Records	XX
7.1	Execution of Aggregate Query	XX
7.2	Execution of UPDATE Operation	XX
7.3	Execution of DELETE Operation	XX
7.4	Execution of Correlated Query	XX
7.5	Execution of Nested Query	XX
8.1	Stored Procedure Creation and Execution	XX
8.2	Function Creation and Execution	XX
8.3	Trigger Creation and Execution	XX
9.1	User Login Page	XX
9.2	Application Dashboard	XX
9.3	Escrow Creation Page	XX
9.4	Multi-Signature Approval Page	XX
9.5	Transaction History Page	XX
9.6	Dispute Management Page	XX

NOTE: all the below sections are single chapters for report so start them in new page and complete all subsection as mentioned.(delete the note once complete report for submission)

1. INTRODUCTION

Digital payments and online transactions have become an essential part of modern commerce. However, transactions between parties who do not necessarily trust each other can create challenges related to payment security, transaction verification and dispute resolution.

In a conventional payment process, buyers are expected to transfer money to sellers before receiving the promised goods or services. This creates risks for both parties. Buyers may be concerned about sellers failing to deliver the agreed product or service, while sellers may be concerned about buyers refusing to complete payment after receiving the service.

An escrow system addresses this problem by introducing a controlled mechanism in which funds are held until predefined transaction conditions are fulfilled. The Decentralized Escrow & Multi-Signature Payment Settlement Engine aims to implement such a mechanism using database management concepts.

The system allows users to register and participate in transactions as buyers, sellers or arbitrators. When an escrow is created, the buyer, seller and applicable multi-signature rule are associated with the escrow record. Authorized participants are added as signers, and their approvals are recorded in the database.

The multi-signature mechanism ensures that a transaction is not released based on a single unauthorized action. Instead, the system checks whether the required number of approvals has been obtained before allowing settlement.

The system also provides mechanisms for handling disputes, refunds and transaction expiry. All major transaction-related information is maintained in a structured database to ensure consistency, integrity and traceability.

MySQL 8 is used as the primary relational database management system. The database consists of seven interconnected tables that represent users, escrow agreements, signing rules, signers, approvals, transactions and disputes. Relational constraints ensure that invalid relationships and inconsistent records are prevented.

A web-based interface provides users with access to the application's features. The project demonstrates the use of database design, relational mapping, SQL queries, aggregate functions, nested queries, correlated queries, stored procedures, functions and triggers.

The primary objective is to develop a reliable database-backed escrow management application that demonstrates how DBMS concepts can be applied to a real-world transaction management problem.

2. PROBLEM DEFINITION

2.1 Problem Statement

In online transactions involving buyers and sellers, there is often a lack of trust between the participating parties. Traditional payment mechanisms may require buyers to transfer funds before receiving the promised goods or services. This creates the possibility of payment fraud, non-delivery, unauthorized transactions and disagreements between parties.

There is a need for a centralized and structured transaction management mechanism that can hold payment information, enforce predefined approval conditions and maintain a reliable record of transaction activities.

The proposed Decentralized Escrow & Multi-Signature Payment Settlement Engine addresses this problem by implementing an escrow management system supported by multi-signature approval rules. The system stores transaction information in a relational database and ensures that settlement operations follow the required approval conditions.

2.2 Existing System

In conventional online transactions:

- Buyers transfer payments directly or through conventional payment gateways.
- Transaction approval may depend on a single party or intermediary.
- Dispute resolution may require manual intervention.
- Transaction records may be distributed across different systems.
- There may be limited visibility into approval and settlement status.

2.3 Proposed System

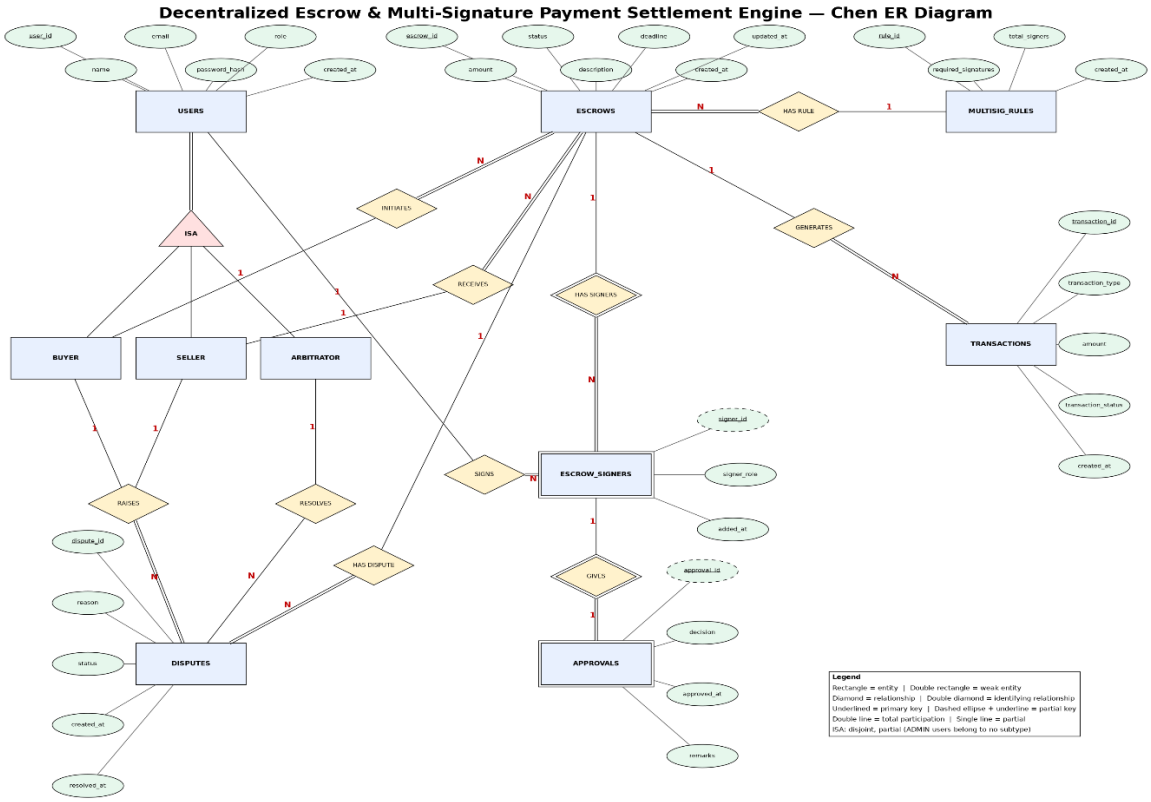
The proposed system introduces an escrow-based transaction management mechanism.

- Users can register as buyers, sellers or arbitrators.
- Buyers can initiate escrow transactions.
- Multi-signature rules determine the number of approvals required.
- Authorized signers can submit transaction approvals.
- Settlement is permitted only when the required approval conditions are satisfied.
- Disputes can be created and managed.
- Transactions can be released, refunded, cancelled or expired based on their lifecycle conditions.
- Database constraints and triggers help maintain consistency.

2.4 Expected Outcomes

The expected outcome is a functional database-driven application that provides structured escrow management, multi-signature approval tracking, transaction monitoring and dispute management while demonstrating important DBMS concepts.

3. ER MODEL



3.1 Entity descriptions

Entity	Description
Users	Stores registered user information and roles.
Multisig Rules	Defines the required and total number of signatures.
Escrows	Stores the details of escrow agreements.
Escrow Signers	Associates authorized signers with escrow agreements.
Approvals	Records approval actions submitted by signers.
Transactions	Maintains settlement and payment transaction records.
Disputes	Stores disputes raised against escrow agreements.

4. ER TO RELATIONAL MAPPING

4.1 STEPS OF ALGORITHM FOR CHOSEN PROBLEM

The Entity Relationship model of the Decentralized Escrow & Multi-Signature Payment Settlement Engine is converted into a relational database schema using standard ER-to-relational mapping rules.

Step 1: Mapping of strong entities

Each strong entity in the ER model is converted into an individual relation. The entities Users, Multisig Rules, Escrows, Escrow Signers, Approvals, Transactions and Disputes are represented as separate relational tables.

Step 2: Mapping of primary keys

A unique primary key is assigned to each relation to identify individual records.

Step 3: Mapping of one-to-many relationships

For one-to-many relationships, the primary key of the one-side relation is included as a foreign key in the many-side relation.

For example, buyer_id and seller_id in the Escrows relation reference the Users relation.

Step 4: Mapping of many-to-many relationships

Many-to-many relationships are represented using an associative relation.

The relationship between Users and Escrows for authorized signers is represented through the Escrow Signers relation.

Step 5: Mapping of relationship attributes

Attributes associated with relationships are stored in their respective associative relations.

For example, approval information is maintained in the Approvals relation.

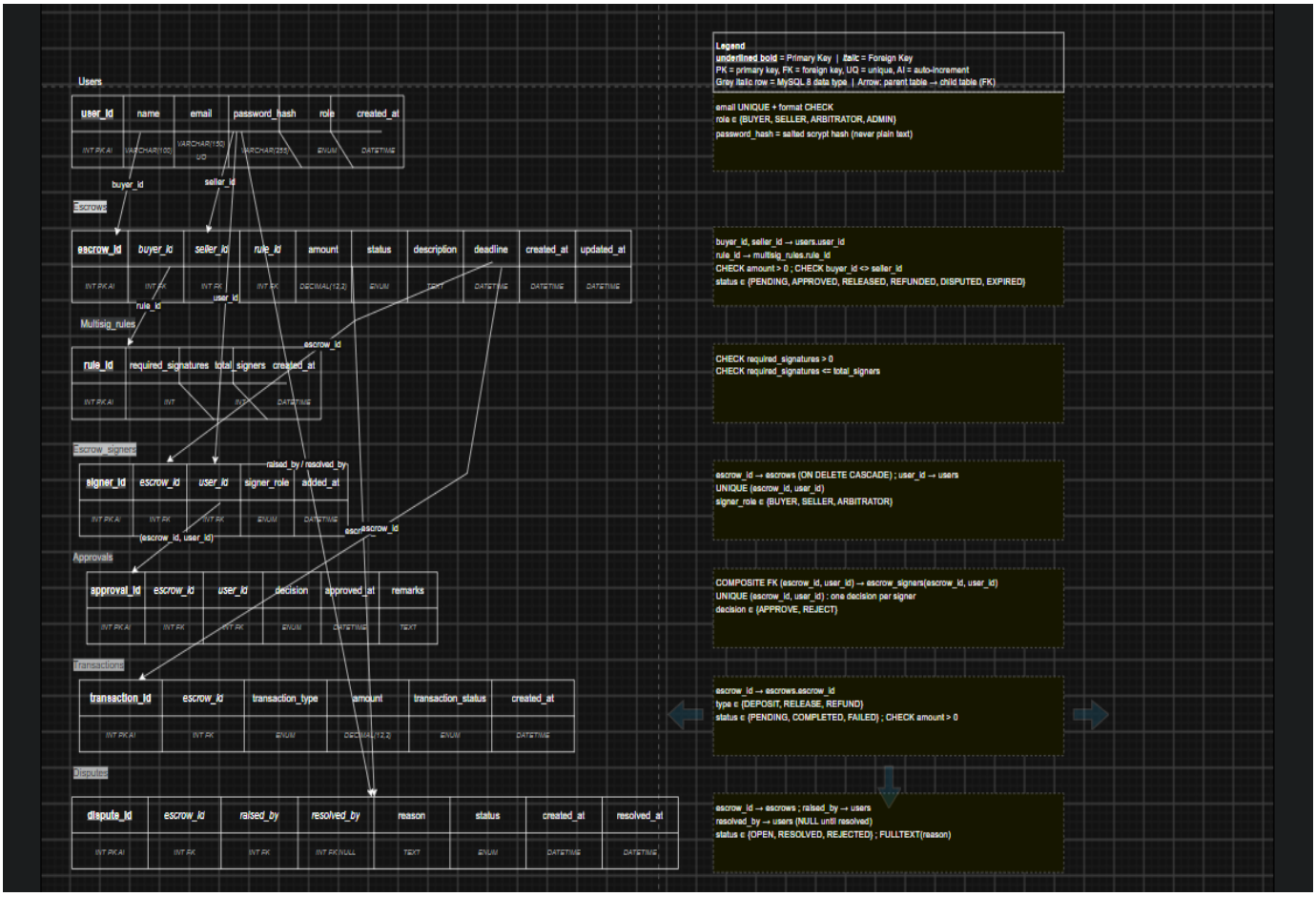
Step 6: Application of integrity constraints

Primary key, foreign key, unique and check constraints are applied to maintain consistency and prevent invalid records.

Step 7: Final relational schema

The resulting relational schema consists of seven interconnected relations that represent the complete database design of the application.

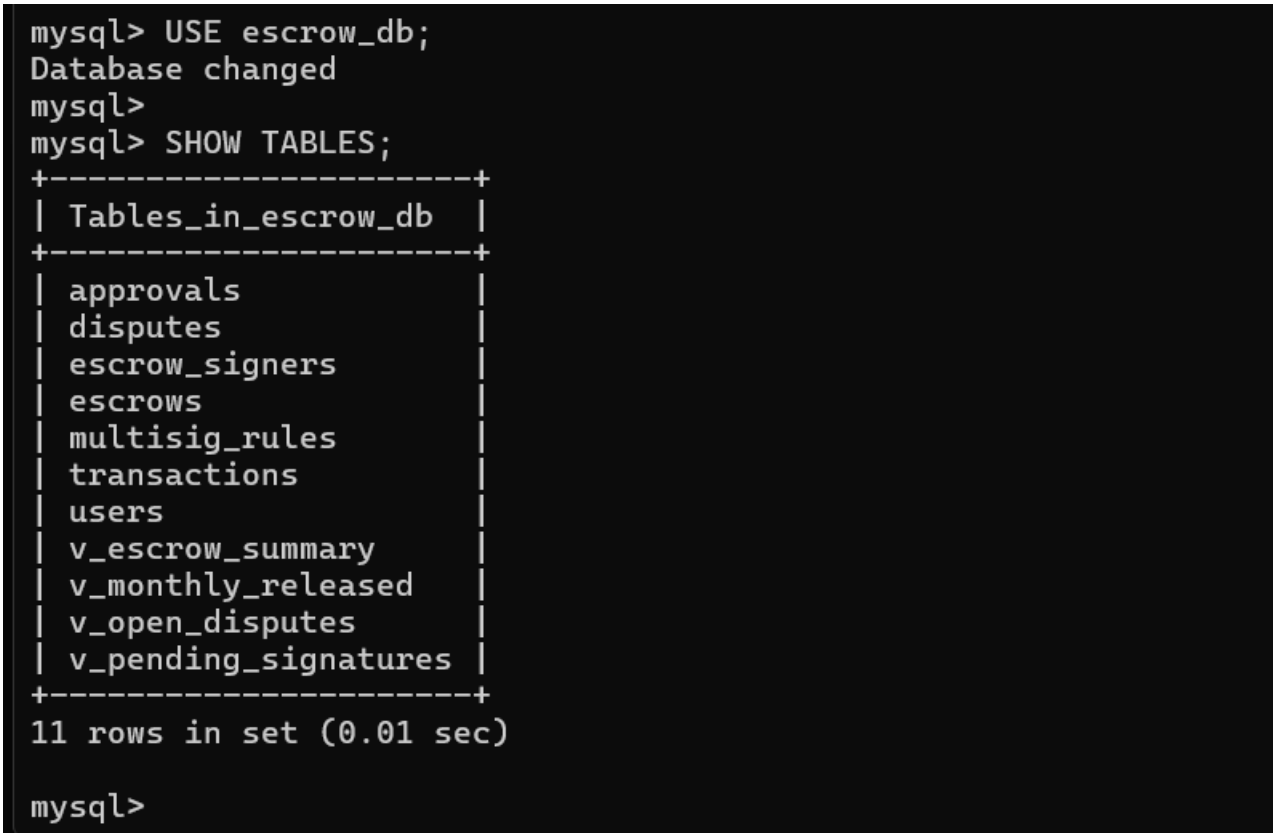
4.2 COMPLETE DIAGRAM OF RELATIONAL MAPPING



5. DDL STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE CREATION

1)All tables created



2) Alter and drop operations

```sql
-- (A) ALTER on real tables: performance + full-text indexes
ALTER TABLE escrows      ADD INDEX idx_escrows_status_deadline (status, deadline);
ALTER TABLE escrows      ADD FULLTEXT INDEX ft_escrows_description (description);
ALTER TABLE disputes     ADD FULLTEXT INDEX ft_disputes_reason (reason);
ALTER TABLE users        ADD INDEX idx_users_role (role);
ALTER TABLE transactions ADD INDEX idx_tx_created (created_at);

-- (B) Temporary index added, then DROPPED
ALTER TABLE users ADD INDEX idx_tmp_name (name);
DROP INDEX idx_tmp_name ON users;

-- (C) Scratch table: CREATE -> ALTER (add/modify/rename/drop column, add/drop CHECK) -> DROP
CREATE TABLE demo_scratch (
  id   INT AUTO_INCREMENT PRIMARY KEY,
  note VARCHAR(50) NULL
);
ALTER TABLE demo_scratch ADD COLUMN flag TINYINT(1) NOT NULL DEFAULT 0;
ALTER TABLE demo_scratch MODIFY COLUMN note VARCHAR(200) NOT NULL DEFAULT 'n/a';
ALTER TABLE demo_scratch ADD CONSTRAINT chk_demo_note CHECK (CHAR_LENGTH(note) > 0);
ALTER TABLE demo_scratch DROP CHECK chk_demo_note;
ALTER TABLE demo_scratch RENAME COLUMN note TO message;
ALTER TABLE demo_scratch DROP COLUMN flag;
DROP TABLE demo_scratch;
```

3) Views

```
mysql> SELECT * FROM v_escrow_summary;
+-----------+---------+---------+-----------------------------------------------------------------+---------------------+---------------------+---------------------+----------+------------+----------+-------------+----------------------+-----------+
| escrow_id | amount  | status  | description                                                     | deadline            | created_at          | updated_at          | buyer_id | buyer_name | seller_id | seller_name | required_signatures | total_si
gners | approvals_count | rejections_count | approval_blocked |
+-----------+---------+---------+-----------------------------------------------------------------+---------------------+---------------------+---------------------+----------+------------+----------+-------------+----------------------+-----------+
|         7 | 15000.00 | PENDING | Quick approval demo escrow (1 of 2)                             | 2026-11-14 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         5 | Arjun Rao   |                    1 |
   2 |               0 |                0 |                0 |
|         3 | 12000.00 | PENDING | Logo and brand kit design                                       | 2026-11-01 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        3 | Rahul Verma |         4 | Priya Nair  |                    2 |
   2 |               0 |                0 |                0 |
|         6 |  8000.00 | EXPIRED | Translation work - overdue demo                                 | 2026-10-24 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         4 | Priya Nair  |                    2 |
   2 |               0 |                0 |                0 |
|        10 |    50.00 | PENDING | tmp                                                             | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
   2 |               0 |                0 |                0 |
|        11 |    77.00 | EXPIRED | Overdue test escrow                                             | 2026-10-24 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   2 |               0 |                0 |                0 |
|        12 |    60.00 | PENDING | Deadline vote test                                              | 2026-10-25 12:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   2 |               0 |                0 |                0 |
|         1 | 25000.00 | APPROVED | Website design - milestone 1 payment                           | 2026-11-08 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   3 |               2 |                0 |                0 |
|         2 | 48000.50 | RELEASED | Second-hand laptop purchase with 6 month warranty              | 2026-11-04 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         5 | Arjun Rao   |                    2 |
   3 |               2 |                0 |                0 |
|         4 | 75000.00 | REFUNDED | Used car delivery to Bengaluru                                 | 2026-10-30 18:32:54 | 2026-10-25 18:32:54 | 2026-10-29 04:04:46 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
   3 |               0 |                0 |                0 |
|         5 |  5000.00 | REFUNDED | Textbook set for semester 5                                    | 2026-10-28 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   3 |               1 |                0 |                0 |
|         8 |  1234.50 | RELEASED | Automated test escrow                                          | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   3 |               2 |                0 |                0 |
|         9 |   999.00 | REFUNDED | Dispute test escrow                                            | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
   3 |               0 |                0 |                0 |
|        13 |  1500.00 | RELEASED | DBMS live demo                                                 | 2026-10-26 12:30:00 | 2026-10-25 13:30:40 | 2026-10-25 13:34:05 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   3 |               2 |                0 |                0 |
|        14 |  1500.00 | REFUNDED | DBMS dispute resolution                                        | 2026-10-26 13:36:00 | 2026-10-25 13:36:53 | 2026-10-25 13:39:12 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
   3 |               0 |                0 |                0 |
+-----------+---------+---------+-----------------------------------------------------------------+---------------------+---------------------+---------------------+----------+------------+----------+-------------+----------------------+-----------+
14 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM v_pending_signatures;
+-----------+---------+-------------+-------------+----------+---------------------+
| escrow_id | user_id | signer_name | signer_role | amount   | deadline            |
+-----------+---------+-------------+-------------+----------+---------------------+
|        12 |       2 | Alice Menon | BUYER       |    60.00 | 2026-10-25 12:14:01 |
|        12 |       4 | Priya Nair  | SELLER      |    60.00 | 2026-10-25 12:14:01 |
|        10 |       3 | Rahul Verma | BUYER       |    50.00 | 2026-10-30 13:14:01 |
|        10 |       5 | Arjun Rao   | SELLER      |    50.00 | 2026-10-30 13:14:01 |
|         3 |       3 | Rahul Verma | BUYER       | 12000.00 | 2026-11-01 18:32:54 |
|         3 |       4 | Priya Nair  | SELLER      | 12000.00 | 2026-11-01 18:32:54 |
|         7 |       2 | Alice Menon | BUYER       | 15000.00 | 2026-11-14 18:32:54 |
|         7 |       5 | Arjun Rao   | SELLER      | 15000.00 | 2026-11-14 18:32:54 |
+-----------+---------+-------------+-------------+----------+---------------------+
8 rows in set (0.01 sec)

mysql>
mysql> SELECT * FROM v_open_disputes;
Empty set (0.01 sec)
```

6. DML STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE WITH INSERTED VALUES

1) Insert and update commands

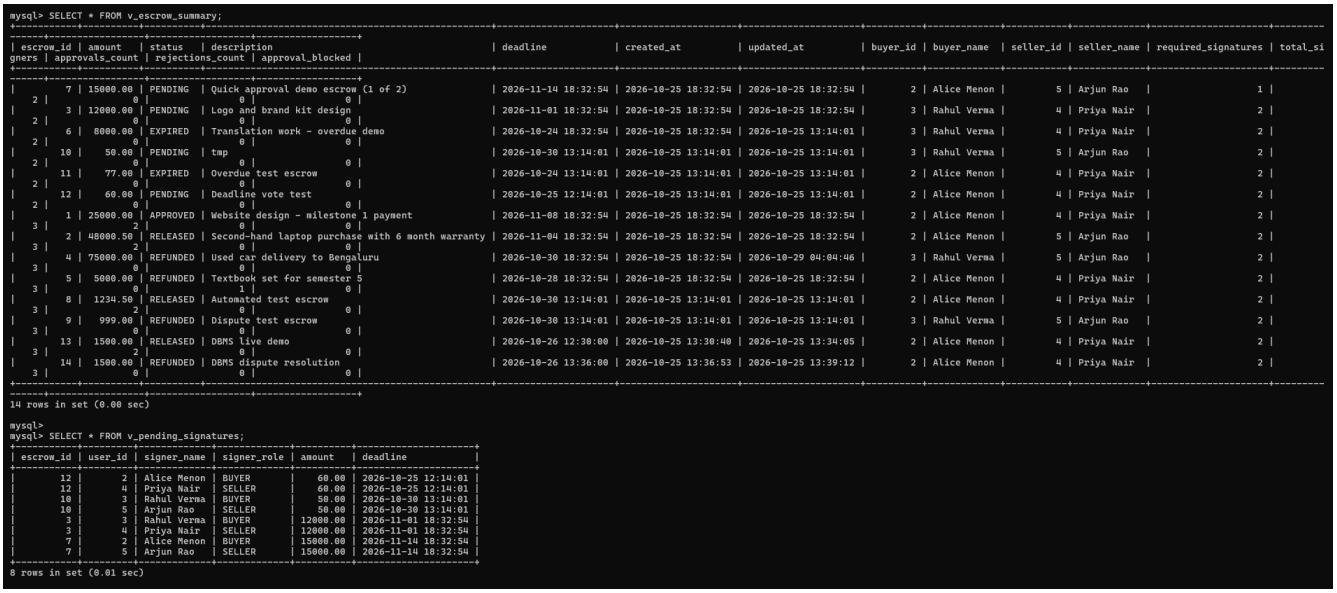
```
-- INSERT
INSERT INTO users (name, email, password_hash, role)
VALUES ('Temp Demo User', 'temp.demo@escrow.test', 'scrypt$demo', 'BUYER');
-- SELECT
SELECT user_id, name, email, role FROM users WHERE email = 'temp.demo@escrow.test';
-- UPDATE
UPDATE users SET name = 'Temp Demo Renamed' WHERE email = 'temp.demo@escrow.test';
-- DELETE
DELETE FROM users WHERE email = 'temp.demo@escrow.test';

-- TRANSACTION + ROLLBACK: a failed multi-step operation leaves NO partial data
SELECT COUNT(*) AS users_before FROM users;
START TRANSACTION;
  INSERT INTO users (name, email, password_hash, role)
  VALUES ('Rollback User', 'rollback.demo@escrow.test', 'scrypt$demo', 'SELLER');
  SELECT COUNT(*) AS users_inside_txn FROM users;      -- +1 inside the transaction
ROLLBACK;
SELECT COUNT(*) AS users_after_rollback FROM users;    -- back to the original number

-- TRANSACTION + COMMIT through a stored procedure (atomic: rule + escrow + signers + deposit)
-- CALL sp_create_escrow(2, 4, 6, 1000.00, 'Demo escrow', DATE_ADD(NOW(), INTERVAL 2 DAY), 2);

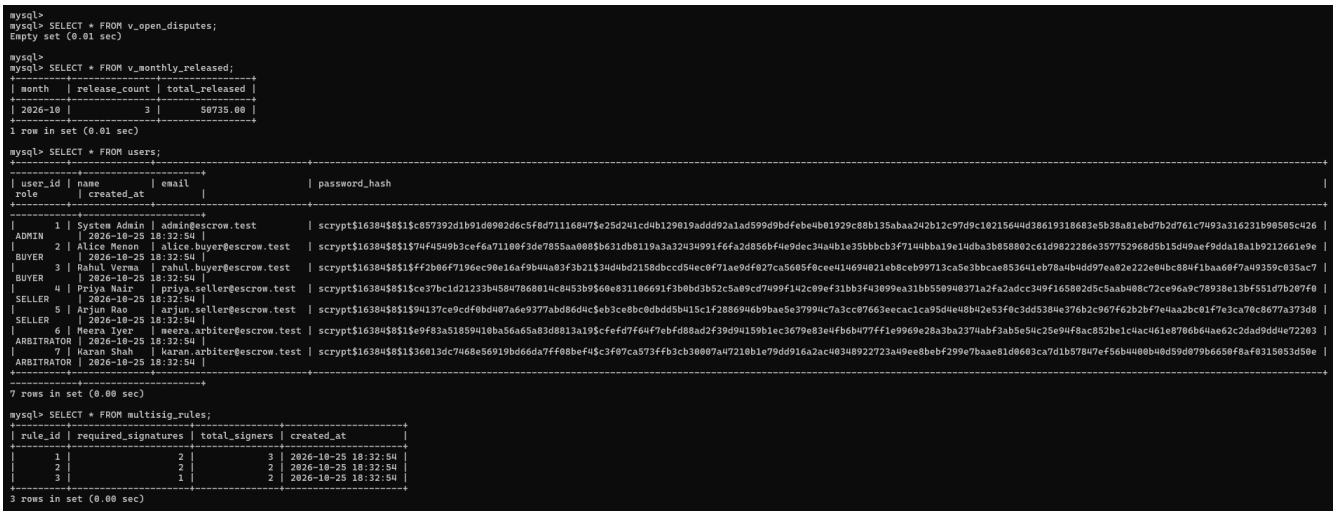
-- ---------------------------------------------------------------------
-- NEGATIVE TESTS (each statement below MUST fail; run one at a time):
-- ---------------------------------------------------------------------
-- INSERT INTO escrows (buyer_id,seller_id,rule_id,amount,deadline) VALUES (2,2,1,100,NOW());   -- CHECK buyer<>seller
-- INSERT INTO escrows (buyer_id,seller_id,rule_id,amount,deadline) VALUES (2,4,1,-5,NOW());    -- CHECK amount>0
-- INSERT INTO users (name,email,password hash,role) VALUES ('X Y','not-an-email','h','BUYER'); -- CHECK email format
```

2)Changes in each table



```
mysql> SELECT * FROM v_escrow_summary;
+-----------+----------+----------+-------------------------------------------------------+---------------------+---------------------+---------------------+----------+-------------+-----------+-------------+----------------------+---------+
| escrow_id | amount   | status   | description                                           | deadline            | created_at          | updated_at          | buyer_id | buyer_name  | seller_id | seller_name | required_signatures | total_si
gners | approvals_count | rejections_count | approval_blocked |
+-----------+----------+----------+-------------------------------------------------------+---------------------+---------------------+---------------------+----------+-------------+-----------+-------------+----------------------+---------
------+-----------------+------------------+------------------+
|         7 | 15000.00 | PENDING  | Quick approval demo escrow (1 of 2)                   | 2026-11-14 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         5 | Arjun Rao   |                    1 |
     2 |               0 |                0 |                0 |
|         3 | 12000.00 | PENDING  | Logo and brand kit design                             | 2026-11-01 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        3 | Rahul Verma |         4 | Priya Nair  |                    2 |
     2 |               0 |                0 |                0 |
|         6 |  8000.00 | EXPIRED  | Translation work - overdue demo                       | 2026-10-24 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         4 | Priya Nair  |                    2 |
     2 |               0 |                0 |                0 |
|        10 |    50.00 | PENDING  | tmp                                                   | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
     2 |               0 |                0 |                0 |
|        11 |    77.00 | EXPIRED  | Overdue test escrow                                   | 2026-10-24 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     2 |               0 |                0 |                0 |
|        12 |    60.00 | PENDING  | Deadline vote test                                    | 2026-10-25 12:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     2 |               0 |                0 |                0 |
|         1 | 25000.00 | APPROVED | Website design - milestone 1 payment                  | 2026-11-08 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     3 |               2 |                0 |                0 |
|         2 | 48000.50 | RELEASED | Second-hand laptop purchase with 6 month warranty     | 2026-11-04 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         5 | Arjun Rao   |                    2 |
     3 |               2 |                0 |                0 |
|         4 | 75000.00 | REFUNDED | Used car delivery to Bengaluru                        | 2026-10-30 18:32:54 | 2026-10-25 18:32:54 | 2026-10-29 04:04:46 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
     3 |               0 |                0 |                0 |
|         5 |  5000.00 | REFUNDED | Textbook set for semester 5                           | 2026-10-28 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     3 |               0 |                1 |                0 |
|         8 |  1234.50 | RELEASED | Automated test escrow                                 | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     3 |               2 |                0 |                0 |
|         9 |   999.00 | REFUNDED | Dispute test escrow                                   | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |        3 | Rahul Verma |         5 | Arjun Rao   |                    2 |
     3 |               0 |                0 |                0 |
|        13 |  1500.00 | RELEASED | DBMS live demo                                        | 2026-10-26 12:30:00 | 2026-10-25 13:30:40 | 2026-10-25 13:34:05 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     3 |               2 |                0 |                0 |
|        14 |  1500.00 | REFUNDED | DBMS dispute resolution                               | 2026-10-26 13:36:00 | 2026-10-25 13:36:53 | 2026-10-25 13:39:12 |        2 | Alice Menon |         4 | Priya Nair  |                    2 |
     3 |               0 |                0 |                0 |
+-----------+----------+----------+-------------------------------------------------------+---------------------+---------------------+---------------------+----------+-------------+-----------+-------------+----------------------+---------
------+-----------------+------------------+------------------+
14 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM v_pending_signatures;
+-----------+---------+-------------+-------------+----------+---------------------+
| escrow_id | user_id | signer_name | signer_role | amount   | deadline            |
+-----------+---------+-------------+-------------+----------+---------------------+
|        12 |       2 | Alice Menon | BUYER       |    60.00 | 2026-10-25 12:14:01 |
|        12 |       4 | Priya Nair  | SELLER      |    60.00 | 2026-10-25 12:14:01 |
|        10 |       3 | Rahul Verma | BUYER       |    50.00 | 2026-10-30 13:14:01 |
|        10 |       5 | Arjun Rao   | SELLER      |    50.00 | 2026-10-30 13:14:01 |
|         3 |       3 | Rahul Verma | BUYER       | 12000.00 | 2026-11-01 18:32:54 |
|         3 |       4 | Priya Nair  | SELLER      | 12000.00 | 2026-11-01 18:32:54 |
|         7 |       2 | Alice Menon | BUYER       | 15000.00 | 2026-11-14 18:32:54 |
|         7 |       5 | Arjun Rao   | SELLER      | 15000.00 | 2026-11-14 18:32:54 |
+-----------+---------+-------------+-------------+----------+---------------------+
8 rows in set (0.01 sec)
```



```
mysql>
mysql> SELECT * FROM v_open_disputes;
Empty set (0.01 sec)

mysql>
mysql> SELECT * FROM v_monthly_released;
+---------+---------------+-----------------+
| month   | release_count | total_released |
+---------+---------------+-----------------+
| 2026-10 |             3 |       50735.00 |
+---------+---------------+-----------------+
1 row in set (0.01 sec)

mysql> SELECT * FROM users;
+---------+-----------------+-----------------------------+--------------------------------------------------------------------------------------------------------------------------------------------------------------+
-----------+---------------------+
| user_id | name            | email                       | password_hash                                                                                                                                                |
 role      | created_at          |
+---------+-----------------+-----------------------------+--------------------------------------------------------------------------------------------------------------------------------------------------------------+
-----------+---------------------+
|       1 | System Admin    | admin@escrow.test           | scrypt$16384$8$1$c857392d1b91d0902d6c5f8d711168475$e25d241cd4b129019addd92a1ad599d9bdfebe4b01929c88b135abaa242b12c97d9c1021564d38619318683e5b38a81ebd7b2d761c7493a316231b90505c426 |
 ADMIN     | 2026-10-25 18:32:54 |
|       2 | Alice Menon     | alice.buyer@escrow.test     | scrypt$16384$8$1$74f4549b3cef6a71100f3de7855aa008$b631db8119a3a32434991f6fa2d856bf4e9dec34a4b1e35bbbcb3f7144bba19e14dba3b858802c61d9822286e357752968d5b15d49aef9dda18a1b9212661e9e |
 BUYER     | 2026-10-25 18:32:54 |
|       3 | Rahul Verma     | rahul.buyer@escrow.test     | scrypt$16384$8$1$ff2b06f7196ec90e16af9b44a03f3b21$34d4bd2158dbccd54ec0f71ae9df027ca5605f0cee411469402 1eb8ceb99713ca5e3bbcae853641eb78a4b4dd97ea02e222e84bc884f1baa60f7a49359c035ac7 |
 BUYER     | 2026-10-25 18:32:54 |
|       4 | Priya Nair      | priya.seller@escrow.test    | scrypt$16384$8$1$ce37bc1d21233b45847868014c8453b9$60e831106691f3b0bd3b52c5a09cd7499f142c09ef31bb3f43099ea31bb550940371a2fa2adcc349f165802d5c5aab408c72ce96a9c78938e13bf551d7b207ff0 |
 SELLER    | 2026-10-25 18:32:54 |
|       5 | Arjun Rao       | arjun.seller@escrow.test    | scrypt$16384$8$1$594137ce9cdf0bd407a6e9377abd86d4c$eb3ce8bc0dbdd5b415c1f2886946b9bae5e37994c7a3cc07663eecac1ca95d4e48b42e53f0c3dd5384e376b2c967f62b2bf7e4aa2bc01f7e3ca70c8677a373d8 |
 SELLER    | 2026-10-25 18:32:54 |
|       6 | Meera Iyer      | meera.arbiter@escrow.test   | scrypt$16384$8$1$e9f83a51859410ba56a65a83d8813a19$cfefd7f64f7ebfd88ad2f39d94159b1ec3679e83e4fb6b477ff1e9969e28a3ba2374abf3ab5e54c25e94f8ac852be1c4ac461e8706b64ae62c2dad9dd4e72203 |
 ARBITRATOR | 2026-10-25 18:32:54 |
|       7 | Karan Shah      | karan.arbiter@escrow.test   | scrypt$16384$8$1$136013dc7468e56919bd66da7ff08bef4$c3f07ca573ffb3cb30007a47210b1e79dd916a2ac4034 8922723a49ee8bebf299e7baae81d0603ca7d1b5784 7ef56b4400b40d59d079b6650f8af0315053d50e |
 ARBITRATOR | 2026-10-25 18:32:54 |
+---------+-----------------+-----------------------------+--------------------------------------------------------------------------------------------------------------------------------------------------------------+
-----------+---------------------+
7 rows in set (0.00 sec)

mysql> SELECT * FROM multisig_rules;
+---------+----------------------+-----------------+---------------------+
| rule_id | required_signatures | total_signers | created_at          |
+---------+----------------------+-----------------+---------------------+
|       1 |                    2 |               3 | 2026-10-25 18:32:54 |
|       2 |                    2 |               2 | 2026-10-25 18:32:54 |
|       3 |                    1 |               2 | 2026-10-25 18:32:54 |
+---------+----------------------+-----------------+---------------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM escrows;
+-----------+----------+-----------+---------+----------+----------+-----------------------------------------------------+---------------------+---------------------+---------------------+
| escrow_id | buyer_id | seller_id | rule_id | amount   | status   | description                                         | deadline            | created_at          | updated_at          |
+-----------+----------+-----------+---------+----------+----------+-----------------------------------------------------+---------------------+---------------------+---------------------+
|         1 |        2 |         4 |       1 | 25000.00 | APPROVED | Website design - milestone 1 payment                | 2026-11-08 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|         2 |        2 |         5 |       1 | 48000.50 | RELEASED | Second-hand laptop purchase with 6 month warranty   | 2026-11-04 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|         3 |        3 |         4 |       2 | 12000.00 | PENDING  | Logo and brand kit design                           | 2026-11-01 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|         4 |        3 |         5 |       1 | 75000.00 | REFUNDED | Used car delivery to Bengaluru                      | 2026-10-30 18:32:54 | 2026-10-25 18:32:54 | 2026-10-29 04:04:46 |
|         5 |        2 |         4 |       1 |  5000.00 | REFUNDED | Textbook set for semester 5                         | 2026-10-28 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|         6 |        3 |         4 |       2 |  8000.00 | EXPIRED  | Translation work - overdue demo                     | 2026-10-24 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 13:14:01 |
|         7 |        2 |         5 |       3 | 15000.00 | PENDING  | Quick approval demo escrow (1 of 2)                 | 2026-11-14 18:32:54 | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|         8 |        2 |         4 |       1 |  1234.50 | RELEASED | Automated test escrow                               | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|         9 |        3 |         5 |       1 |   999.00 | REFUNDED | Dispute test escrow                                 | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|        10 |        3 |         5 |       2 |    50.00 | PENDING  | tmp                                                 | 2026-10-30 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|        11 |        2 |         4 |       2 |    77.00 | EXPIRED  | Overdue test escrow                                 | 2026-10-24 13:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|        12 |        2 |         4 |       2 |    60.00 | PENDING  | Deadline vote test                                  | 2026-10-25 12:14:01 | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|        13 |        2 |         4 |       1 |  1500.00 | RELEASED | DBMS live demo                                      | 2026-10-26 12:30:00 | 2026-10-25 13:30:40 | 2026-10-25 13:34:05 |
|        14 |        2 |         4 |       1 |  1500.00 | REFUNDED | DBMS dispute resolution                             | 2026-10-26 13:36:00 | 2026-10-25 13:36:53 | 2026-10-25 13:39:12 |
+-----------+----------+-----------+---------+----------+----------+-----------------------------------------------------+---------------------+---------------------+---------------------+
14 rows in set (0.00 sec)

mysql> SELECT * FROM escrow_signers;
+-----------+-----------+---------+-------------+---------------------+
| signer_id | escrow_id | user_id | signer_role | added_at            |
+-----------+-----------+---------+-------------+---------------------+
|         1 |         1 |       2 | BUYER       | 2026-10-25 18:32:54 |
|         2 |         1 |       4 | SELLER      | 2026-10-25 18:32:54 |
|         3 |         1 |       6 | ARBITRATOR  | 2026-10-25 18:32:54 |
|         4 |         2 |       2 | BUYER       | 2026-10-25 18:32:54 |
|         5 |         2 |       5 | SELLER      | 2026-10-25 18:32:54 |
|         6 |         2 |       7 | ARBITRATOR  | 2026-10-25 18:32:54 |
|         7 |         3 |       3 | BUYER       | 2026-10-25 18:32:54 |
|         8 |         3 |       4 | SELLER      | 2026-10-25 18:32:54 |
|         9 |         4 |       3 | BUYER       | 2026-10-25 18:32:54 |
|        10 |         4 |       5 | SELLER      | 2026-10-25 18:32:54 |
|        11 |         4 |       6 | ARBITRATOR  | 2026-10-25 18:32:54 |
|        12 |         5 |       2 | BUYER       | 2026-10-25 18:32:54 |
|        13 |         5 |       4 | SELLER      | 2026-10-25 18:32:54 |
|        14 |         5 |       7 | ARBITRATOR  | 2026-10-25 18:32:54 |
|        15 |         6 |       3 | BUYER       | 2026-10-25 18:32:54 |
|        16 |         6 |       4 | SELLER      | 2026-10-25 18:32:54 |
|        17 |         7 |       2 | BUYER       | 2026-10-25 18:32:54 |
|        18 |         7 |       5 | SELLER      | 2026-10-25 18:32:54 |
|        19 |         8 |       2 | BUYER       | 2026-10-25 13:14:01 |
|        20 |         8 |       4 | SELLER      | 2026-10-25 13:14:01 |
|        21 |         8 |       6 | ARBITRATOR  | 2026-10-25 13:14:01 |
|        22 |         9 |       3 | BUYER       | 2026-10-25 13:14:01 |
|        23 |         9 |       5 | SELLER      | 2026-10-25 13:14:01 |
|        24 |         9 |       7 | ARBITRATOR  | 2026-10-25 13:14:01 |
|        25 |        10 |       3 | BUYER       | 2026-10-25 13:14:01 |
|        26 |        10 |       5 | SELLER      | 2026-10-25 13:14:01 |
|        27 |        11 |       2 | BUYER       | 2026-10-25 13:14:01 |
|        28 |        11 |       4 | SELLER      | 2026-10-25 13:14:01 |
|        29 |        12 |       2 | BUYER       | 2026-10-25 13:14:01 |
|        30 |        12 |       4 | SELLER      | 2026-10-25 13:14:01 |
|        31 |        13 |       2 | BUYER       | 2026-10-25 13:30:40 |
|        32 |        13 |       4 | SELLER      | 2026-10-25 13:30:40 |
|        33 |        13 |       6 | ARBITRATOR  | 2026-10-25 13:30:40 |
|        34 |        14 |       2 | BUYER       | 2026-10-25 13:36:53 |
|        35 |        14 |       4 | SELLER      | 2026-10-25 13:36:53 |
|        36 |        14 |       6 | ARBITRATOR  | 2026-10-25 13:36:53 |
+-----------+-----------+---------+-------------+---------------------+
```

```
mysql> SELECT * FROM approvals;
+-------------+-----------+---------+----------+---------------------+---------------------------------+
| approval_id | escrow_id | user_id | decision | approved_at         | remarks                         |
+-------------+-----------+---------+----------+---------------------+---------------------------------+
|           1 |         1 |       2 | APPROVE  | 2026-10-25 18:32:54 | Design files received           |
|           2 |         1 |       4 | APPROVE  | 2026-10-25 18:32:54 | Delivered as agreed             |
|           3 |         2 |       2 | APPROVE  | 2026-10-25 18:32:54 | Laptop works fine               |
|           4 |         2 |       5 | APPROVE  | 2026-10-25 18:32:54 | NULL                            |
|           5 |         5 |       2 | REJECT   | 2026-10-25 18:32:54 | Wrong edition received          |
|           6 |         8 |       2 | APPROVE  | 2026-10-25 13:14:01 | ok                              |
|           8 |         8 |       4 | APPROVE  | 2026-10-25 13:14:01 | NULL                            |
|           9 |        13 |       2 | APPROVE  | 2026-10-25 13:32:01 | Buyer confirms the milestone terms |
|          10 |        13 |       4 | APPROVE  | 2026-10-25 13:33:40 | NULL                            |
+-------------+-----------+---------+----------+---------------------+---------------------------------+
9 rows in set (0.00 sec)

mysql> SELECT * FROM transactions;
+----------------+-----------+--------------------+----------+--------------------+---------------------+
| transaction_id | escrow_id | transaction_type   | amount   | transaction_status | created_at          |
+----------------+-----------+--------------------+----------+--------------------+---------------------+
|              1 |         1 | DEPOSIT            | 25000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              2 |         2 | DEPOSIT            | 48000.50 | COMPLETED          | 2026-10-25 18:32:54 |
|              3 |         2 | RELEASE            | 48000.50 | COMPLETED          | 2026-10-25 18:32:54 |
|              4 |         3 | DEPOSIT            | 12000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              5 |         4 | DEPOSIT            | 75000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              6 |         5 | DEPOSIT            |  5000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              7 |         5 | REFUND             |  5000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              8 |         6 | DEPOSIT            |  8000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|              9 |         7 | DEPOSIT            | 15000.00 | COMPLETED          | 2026-10-25 18:32:54 |
|             10 |         8 | DEPOSIT            |  1234.50 | COMPLETED          | 2026-10-25 13:14:01 |
|             11 |         8 | RELEASE            |  1234.50 | COMPLETED          | 2026-10-25 13:14:01 |
|             12 |         9 | DEPOSIT            |   999.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             13 |        10 | DEPOSIT            |    50.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             14 |         9 | REFUND             |   999.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             15 |        11 | DEPOSIT            |    77.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             16 |        11 | REFUND             |    77.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             17 |         6 | REFUND             |  8000.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             18 |        12 | DEPOSIT            |    60.00 | COMPLETED          | 2026-10-25 13:14:01 |
|             19 |        13 | DEPOSIT            |  1500.00 | COMPLETED          | 2026-10-25 13:30:40 |
|             20 |        13 | RELEASE            |  1500.00 | COMPLETED          | 2026-10-25 13:34:05 |
|             21 |        14 | DEPOSIT            |  1500.00 | COMPLETED          | 2026-10-25 13:36:53 |
|             22 |        14 | REFUND             |  1500.00 | COMPLETED          | 2026-10-25 13:39:12 |
|             23 |         4 | REFUND             | 75000.00 | COMPLETED          | 2026-10-29 04:04:46 |
+----------------+-----------+--------------------+----------+--------------------+---------------------+
23 rows in set (0.00 sec)

mysql> SELECT * FROM disputes;
+------------+-----------+-----------+-------------+-----------------------------------------------------------------------------------------------------------+----------+---------------------+---------------------+
| dispute_id | escrow_id | raised_by | resolved_by | reason                                                                                                    | status   | created_at          | resolved_at         |
+------------+-----------+-----------+-------------+-----------------------------------------------------------------------------------------------------------+----------+---------------------+---------------------+
|          1 |         4 |         3 |           6 | Seller has not delivered the car within the agreed delivery window and is not responding.                 | RESOLVED | 2026-10-25 18:32:54 | 2026-10-29 04:04:46 |
|          2 |         5 |         2 |           7 | Textbook edition delivered does not match the listing, wrong edition received.                            | RESOLVED | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 |
|          3 |         9 |         3 |           7 | Item never arrived, seller silent for days                                                                | RESOLVED | 2026-10-25 13:14:01 | 2026-10-25 13:14:01 |
|          4 |        14 |         2 |           6 | The seller is not trusted                                                                                 | RESOLVED | 2026-10-25 13:37:38 | 2026-10-25 13:39:12 |
+------------+-----------+-----------+-------------+-----------------------------------------------------------------------------------------------------------+----------+---------------------+---------------------+
4 rows in set (0.00 sec)

mysql>
```

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGRREGATE

1) Group escrow amounts by status

```
mysql> SELECT
    ->     status,
    ->     COUNT(*) AS escrows,
    ->     SUM(amount) AS total_amount,
    ->     AVG(amount) AS avg_amount
    -> FROM escrows
    -> GROUP BY status
    -> ORDER BY total_amount DESC;
+----------+---------+--------------+--------------+
| status   | escrows | total_amount | avg_amount   |
+----------+---------+--------------+--------------+
| REFUNDED |       4 |     82499.00 | 20624.750000 |
| RELEASED |       3 |     50735.00 | 16911.666667 |
| PENDING  |       4 |     27110.00 |  6777.500000 |
| APPROVED |       1 |     25000.00 | 25000.000000 |
| EXPIRED  |       2 |      8077.00 |  4038.500000 |
+----------+---------+--------------+--------------+
5 rows in set (0.00 sec)

mysql>
```

2) Buyers who have locked more than ₹20,000

```
mysql> SELECT
    ->     u.name,
    ->     COUNT(*) AS escrows,
    ->     SUM(e.amount) AS total_locked
    -> FROM escrows e
    -> JOIN users u
    ->     ON u.user_id = e.buyer_id
    -> GROUP BY u.user_id, u.name
    -> HAVING SUM(e.amount) > 20000
    -> ORDER BY total_locked DESC;
+-------------+---------+--------------+
| name        | escrows | total_locked |
+-------------+---------+--------------+
| Alice Menon |       9 |     97372.00 |
| Rahul Verma |       5 |     96049.00 |
+-------------+---------+--------------+
2 rows in set (0.00 sec)

mysql>
```

7.2 CORRELATED QUERY

Find sellers who have never received a RELEASE payout.

```
mysql> SELECT
    ->     u.user_id,
    ->     u.name
    -> FROM users u
    -> WHERE u.role = 'SELLER'
    -> AND NOT EXISTS (
    ->     SELECT 1
    ->     FROM escrows e
    ->     JOIN transactions t
    ->         ON t.escrow_id = e.escrow_id
    ->     WHERE e.seller_id = u.user_id
    ->       AND t.transaction_type = 'RELEASE'
    -> );
Empty set (0.00 sec)
```

7.3 NESTED QUERY

```
mysql> SELECT
    ->     escrow_id,
    ->     buyer_id,
    ->     seller_id,
    ->     amount,
    ->     status
    -> FROM escrows
    -> WHERE amount > (
    ->     SELECT AVG(amount)
    ->     FROM escrows
    -> );
+-----------+----------+-----------+----------+----------+
| escrow_id | buyer_id | seller_id | amount   | status   |
+-----------+----------+-----------+----------+----------+
|         1 |        2 |         4 | 25000.00 | APPROVED |
|         2 |        2 |         5 | 48000.50 | RELEASED |
|         4 |        3 |         5 | 75000.00 | REFUNDED |
|         7 |        2 |         5 | 15000.00 | PENDING  |
+-----------+----------+-----------+----------+----------+
4 rows in set (0.00 sec)

mysql>
```

8. STORED PROCEDURES, FUCNTIONS AND TRIGGERS

8.1 STORED PROCEDURES OR FUNCTIONS

Stored procedures

```
mysql> SHOW PROCEDURE STATUS
    -> WHERE Db = 'escrow_db';
+-----------+--------------------------+-----------+----------------+---------------------+---------------------+---------------+---------+----------------
------+----------------------+--------------------+
| Db        | Name                     | Type      | Definer        | Modified            | Created             | Security_type | Comment | character_set_c
lient | collation_connection | Database Collation |
+-----------+--------------------------+-----------+----------------+---------------------+---------------------+---------------+---------+----------------
------+----------------------+--------------------+
| escrow_db | sp_approve_escrow        | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
| escrow_db | sp_create_escrow         | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
| escrow_db | sp_expire_overdue_escrows | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
| escrow_db | sp_raise_dispute         | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
| escrow_db | sp_release_escrow        | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
| escrow_db | sp_resolve_dispute       | PROCEDURE | root@localhost | 2026-10-25 18:32:54 | 2026-10-25 18:32:54 | DEFINER       |         | cp850
      | cp850_general_ci     | utf8mb4_0900_ai_ci |
+-----------+--------------------------+-----------+----------------+---------------------+---------------------+---------------+---------+----------------
------+----------------------+--------------------+
6 rows in set (0.01 sec)

mysql>
```

8.2 TRIGGERS

Demo of trg_approvals_after_insert

```
mysql> UPDATE escrows
    -> SET amount = 100
    -> WHERE escrow_id = 1;
ERROR 1644 (45000): Escrow amount is immutable
mysql>
```

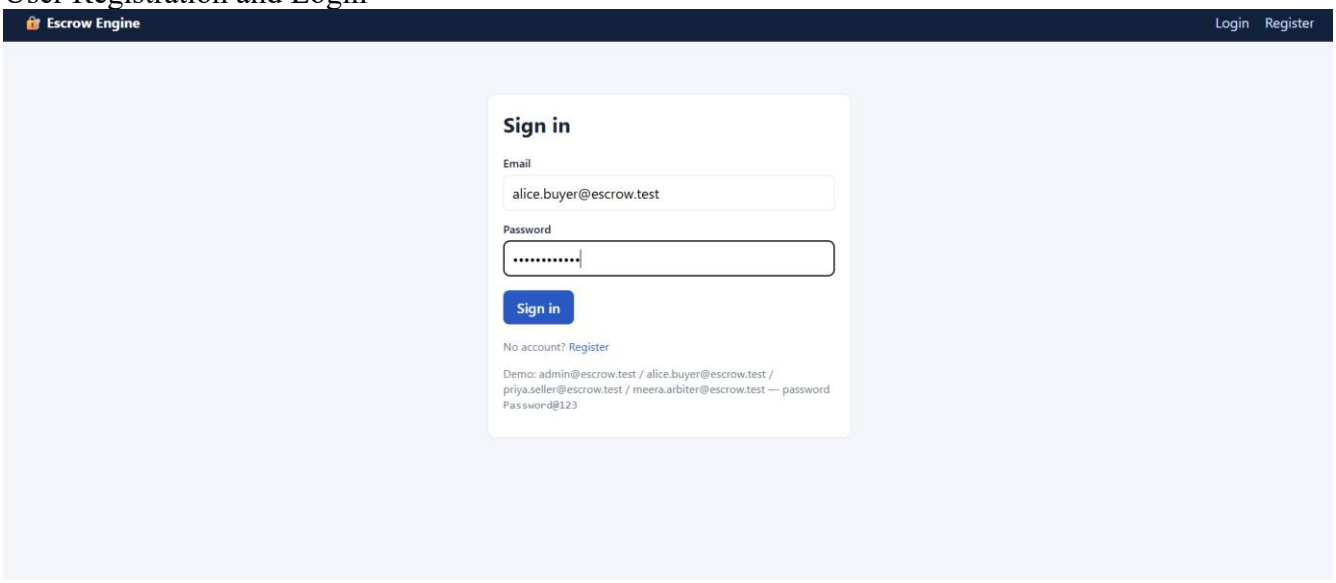
```
mysql> DELETE FROM transactions
    -> WHERE transaction_id = 1;
ERROR 1644 (45000): Ledger entries cannot be deleted
mysql>
```

9. FRONT END DEVELOPEMNT

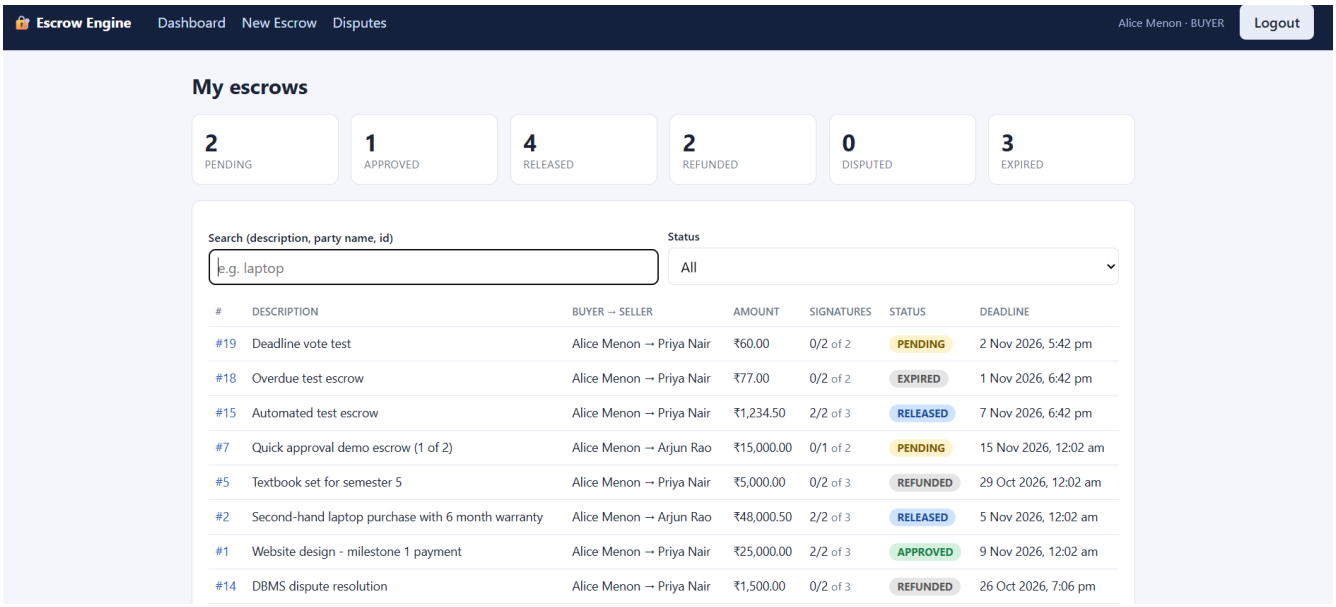
The frontend of the Decentralized Escrow & Multi-Signature Payment Settlement Engine is developed using Next.js, providing a web-based interface for interacting with the escrow management system.
The frontend allows users to access the application's functionality based on their roles. Users can register, log in and view their relevant transaction information.
The application provides interfaces for creating escrow agreements, viewing escrow details, managing multi-signature approvals and monitoring transaction status.
The system also provides functionality for viewing and managing disputes. Users can access transaction information and monitor the approval progress associated with an escrow agreement.
The frontend communicates with the backend application through API endpoints. The backend performs validation and executes the necessary database operations using MySQL.
The application uses role-based access control to restrict access to relevant operations. Buyer, Seller and Arbitrator roles have different responsibilities within the escrow lifecycle.
The frontend and backend together provide an integrated application in which database records are created, updated and retrieved through user interactions.

9.1 Main frontend modules

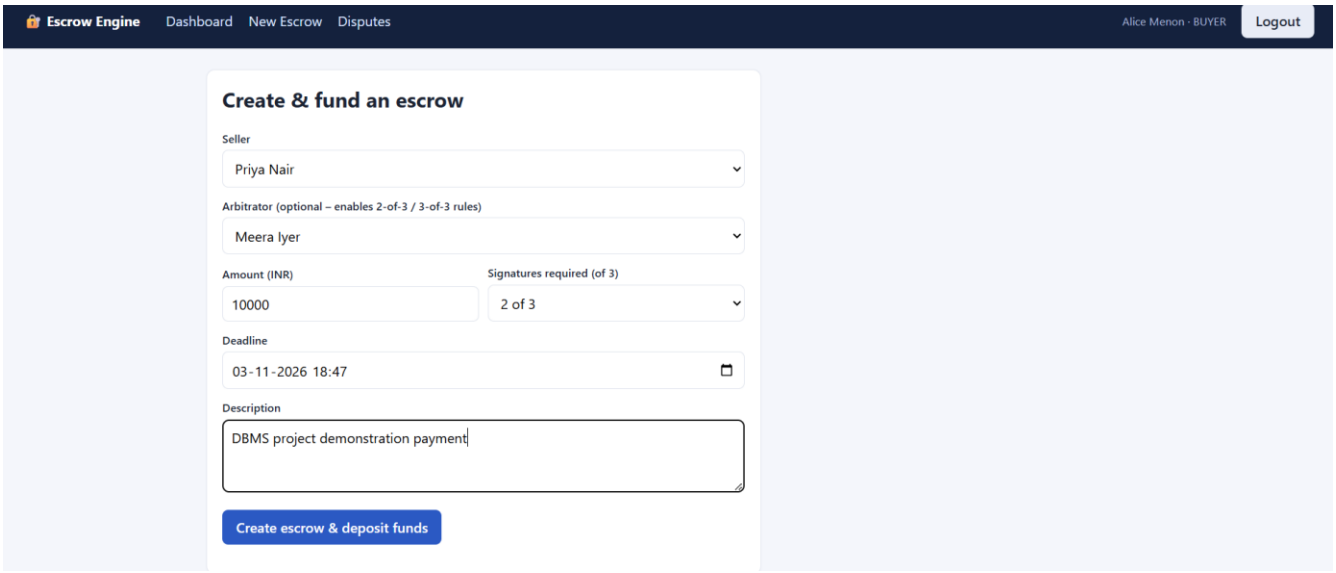
1. User Registration and Login



2. Dashboard



3. Escrow Creation and Management



Escrow #20 PENDING

DBMS project demonstration payment

Amount	Buyer	Seller	Deadline
₹10,000.00	Alice Menon	Priya Nair	3 Nov 2026, 6:47 pm

Multi-signature progress: **0 / 2** approvals required (2-of-3 rule)

Signers

NAME	ROLE	DECISION	WHEN	REMARKS
Alice Menon	BUYER	not voted	-	
Priya Nair	SELLER	not voted	-	
Meera Iyer	ARBITRATOR	not voted	-	

Your remarks (optional)

Approve release | Reject

Raise a dispute

Describe the problem (min 10 characters)

Raise dispute

Ledger (transactions)

#	TYPE	AMOUNT	STATUS	DATE
35	DEPOSIT	₹10,000.00	COMPLETED	2 Nov 2026, 6:47 pm

Disputes

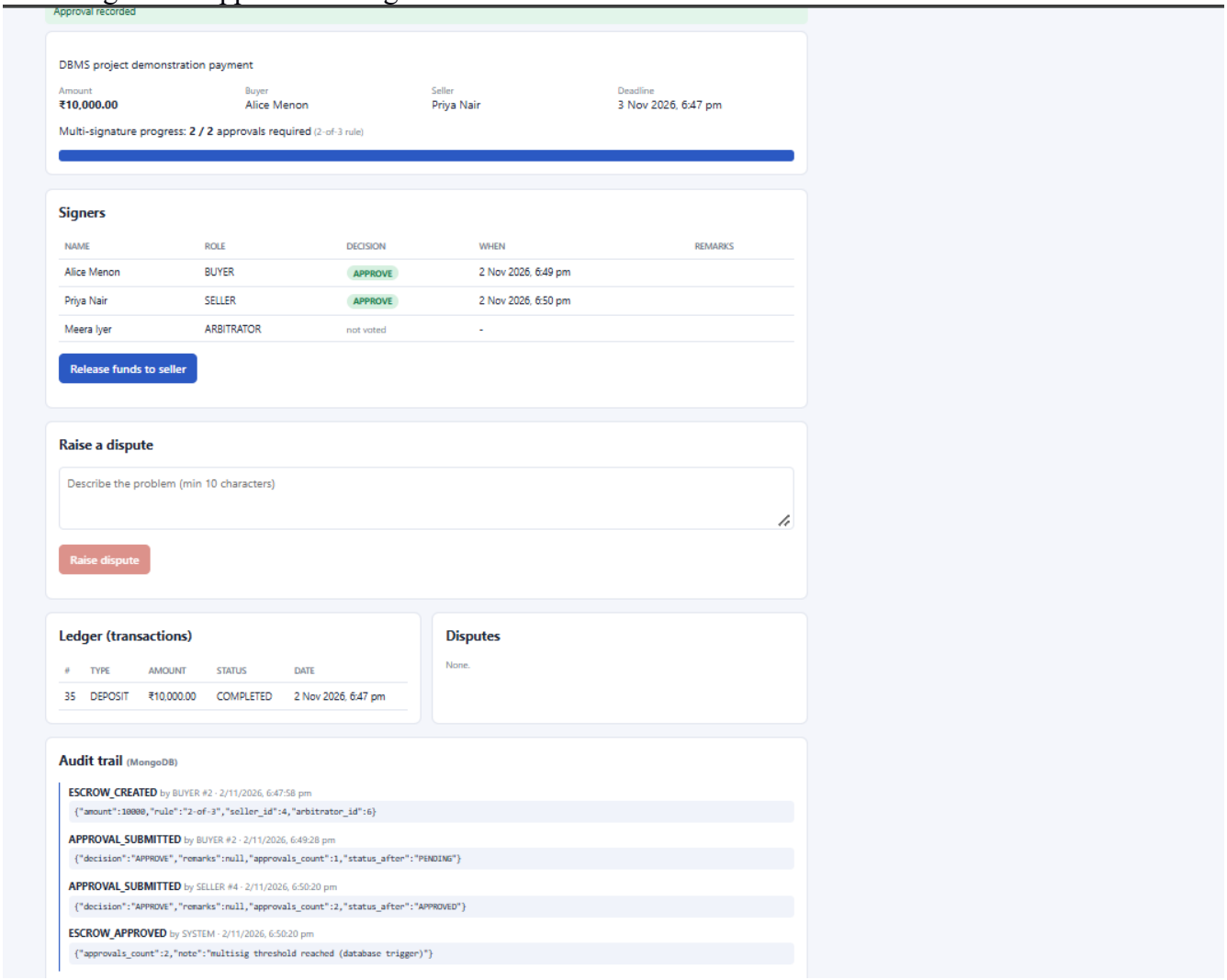
None.

Audit trail (MongoDB)

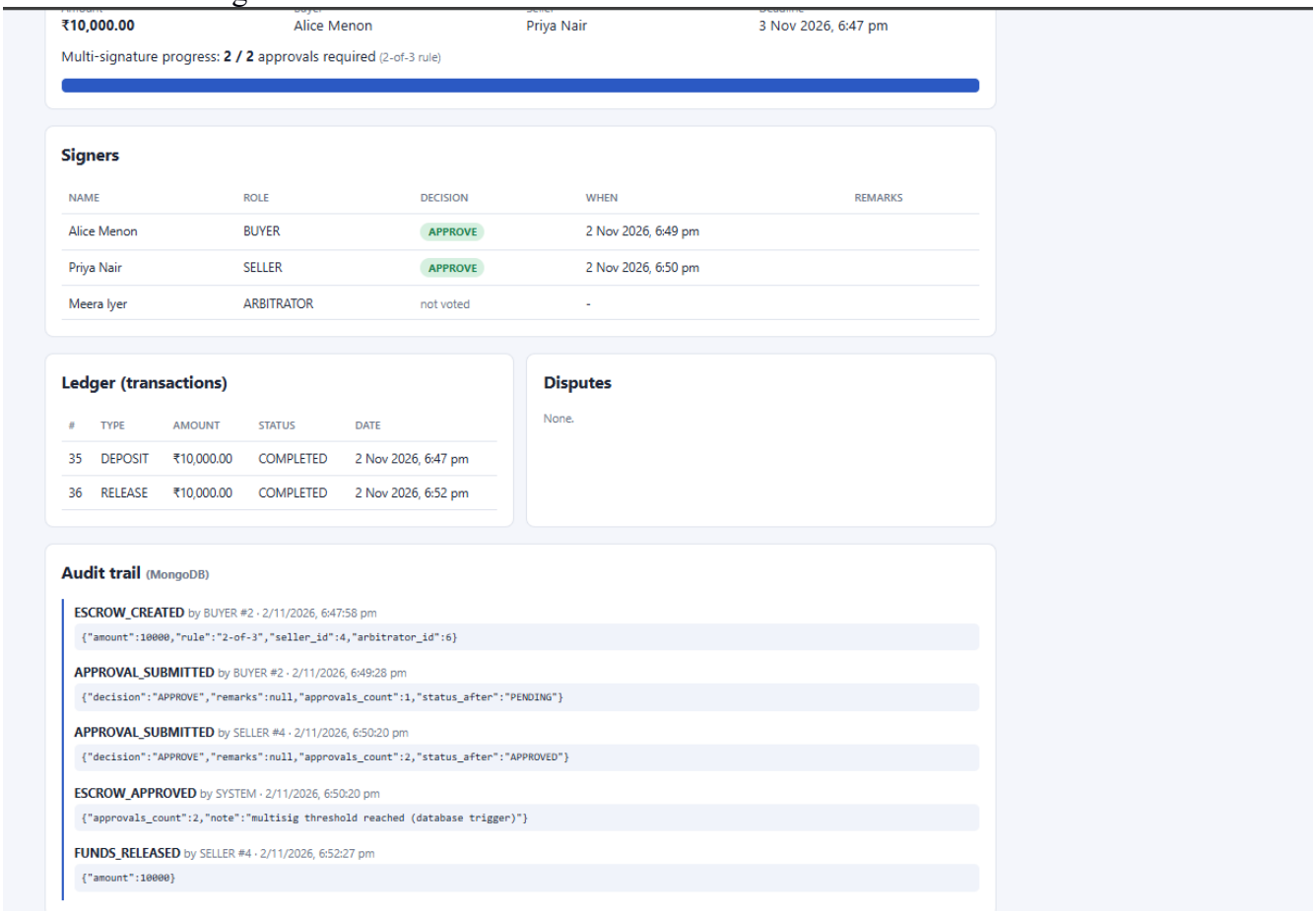
ESCROW_CREATED by BUYER #2 · 2/11/2026, 6:47:58 pm

{"amount":10000,"rule":"2-of-3","seller_id":4,"arbitrator_id":6}

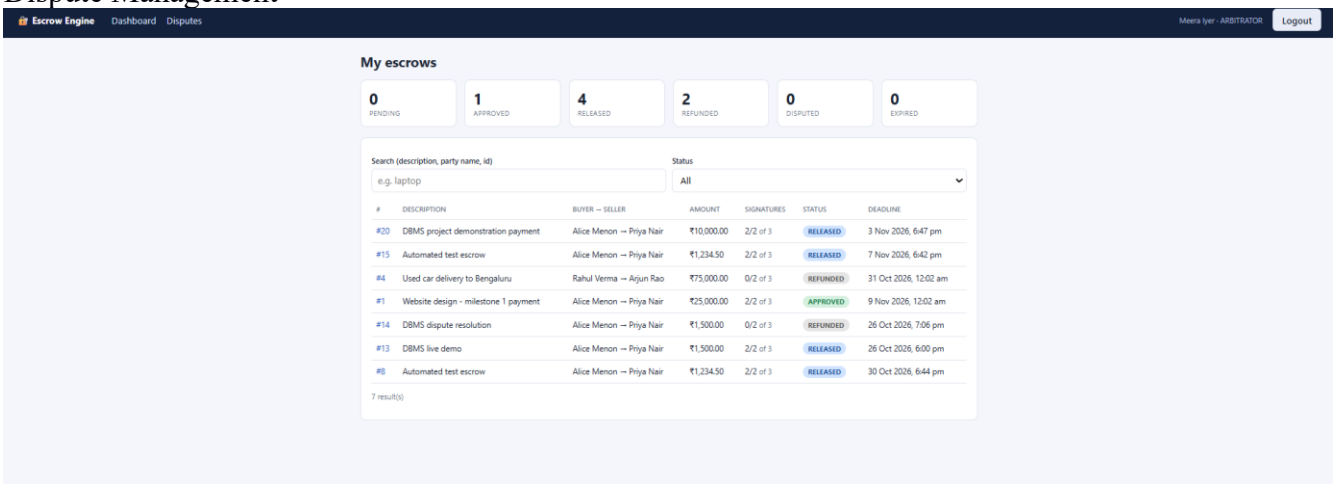
4. Multi-Signature Approval Management

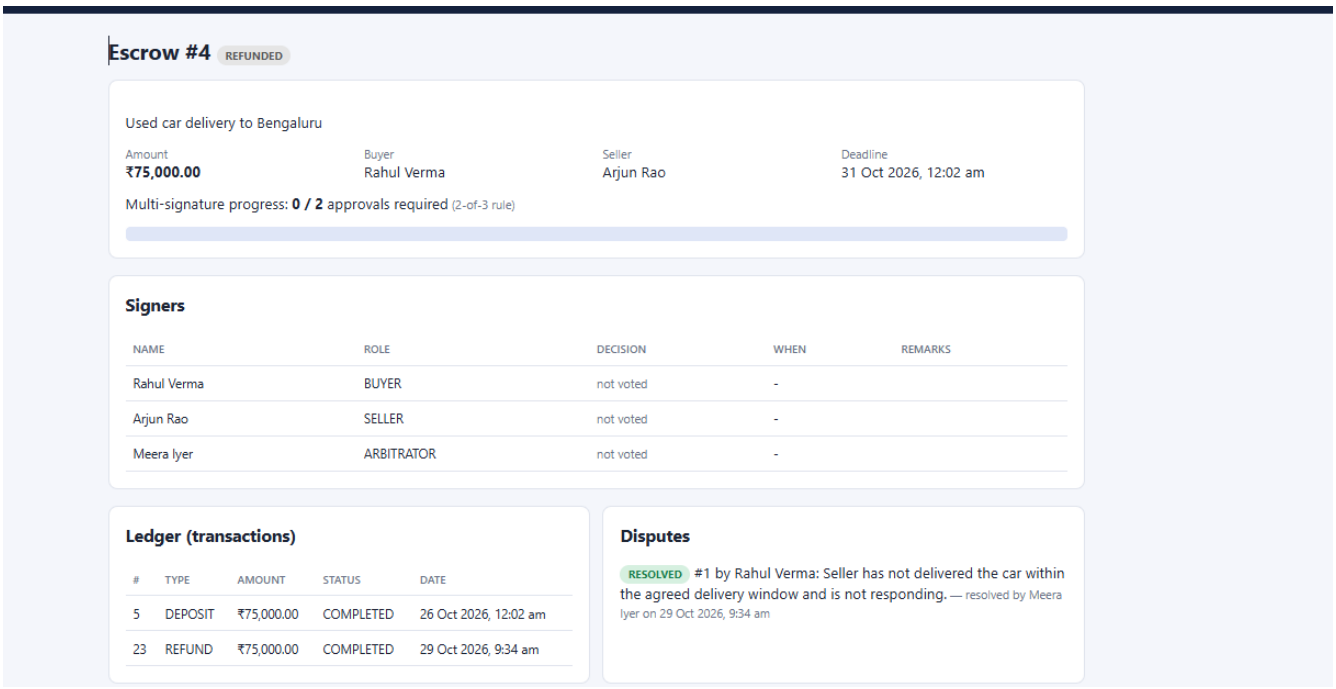


5. Transaction Management



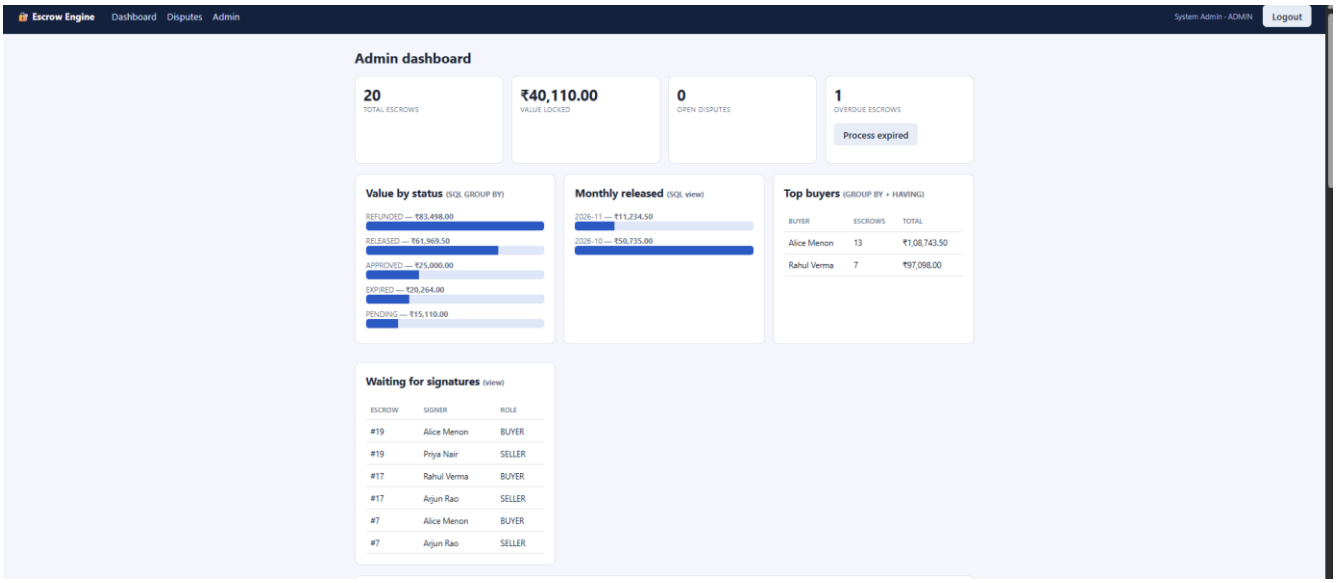
6. Dispute Management



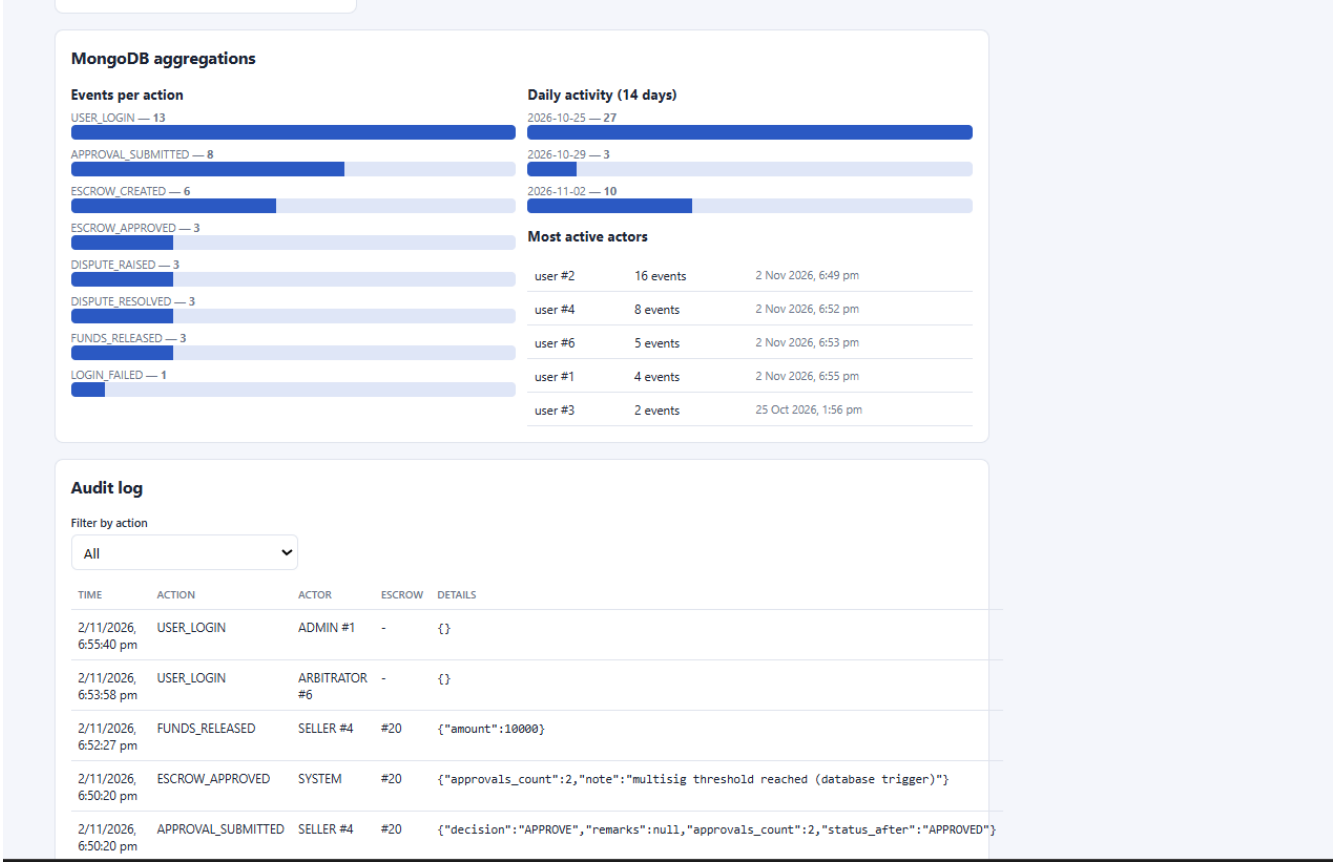


7. User and Role Management

Admin dashboard



Mongodb audit logs



9.2 Database Integration

MySQL 8 serves as the primary relational database. The backend communicates with MySQL to perform database operations such as insertion, retrieval, updating and deletion of records.
Database constraints and validation mechanisms help ensure that the information stored remains consistent and valid.
If MongoDB audit logging is implemented, it provides an additional mechanism for maintaining system activity records.

9.3 Advantages

- User-friendly web interface.
- Role-based application access.
- Centralized transaction management.
- Approval tracking.
- Structured dispute management.
- Database-backed transaction records.

REFERENCES

[1] Oracle, "MySQL 8.0 Reference Manual." Available: https://dev.mysql.com/doc/refman/8.0/en/

[2] MongoDB Inc., "MongoDB Manual." Available: https://www.mongodb.com/docs/manual/

GitHub Link:

https://github.com/Murali-R-123/Escrow-and-Multi-Signature-Payment-Settlement-System